

MEMORANDUM

TO: Festival Operations Manager, Bosphorus Drone Light Festival
FROM: Team Mehmet — IE 306 Assignment 3
DATE: 25 May 2026
SUBJECT: Recommendation: Purchase of One Additional Battery-Swap Station (Policy B)

Executive Summary

We recommend the immediate purchase of one additional battery-swap station (Policy B: 6 stations total). A rigorous steady-state discrete-event simulation with Common Random Numbers (CRN) across 40 replications shows that Policy B reduces mean per-drone swap-queue waiting time by **10.41 seconds** (95% CI: [9.96, 10.85] s). The entire confidence interval lies strictly above zero, and the improvement is both statistically significant and operationally meaningful for continuous festival operations.

Background and Model Overview

The system consists of a closed fleet of 200 drones cycling continuously between the flight pool and the depot. Upon landing, each drone joins the swap queue, proceeds to one of the battery-swap stations, passes through a single test rig, and returns to the flight pool. Because the fleet operates 24/7 with no planned idle periods, the system is modeled as **steady-state**.

Output Analysis Methodology

Steady-State Determination

A conservative warmup period of 10 000 s (≈ 2.8 hours) was deleted from every replication. This value exceeds both Welch's graphical method ($\approx 8,820$ s) and the MSER estimate ($\approx 3,329$ s), ensuring negligible initial transient bias.

Statistical Design

Two complementary steady-state estimation methods were employed:

- **Replications-with-deletion:** 40 independent replications, each of 12 h simulated time.
- **Batch means:** A single long run of 30 h divided into 30 batches of 1 h after warmup deletion.

All random streams (arrivals, service times, routing) used documented NumPy multi-stream seeding to guarantee reproducibility and enable CRN.

Key Results

Table 1: Steady-State Performance Comparison (95% CI)

Metric	Policy A (5 stations)	Policy B (6 stations)
Mean swap-queue wait (s)	14.52 [13.97, 15.07]	4.11 [3.89, 4.33]
Half-width (% of mean)	3.8%	5.4%

CRN Paired Comparison ($A - B$):

- Point estimate of difference: **10.41 s**
- 95% confidence interval: [9.96, 10.85] s
- Variance Reduction Factor (VRF): **1.75**

The CRN design produced a positive correlation of approximately 0.43 between the two policies, reducing the variance of the difference by 43% relative to independent sampling. Because the entire 95% CI lies well above zero, we conclude with high confidence that Policy B reduces average depot waiting time by 10–11 seconds per visit.

Verification and Validation

The model passed two critical V&V checks:

1. **Fleet Conservation Invariant:** The identity

$$\text{FLEET_SIZE} = n_{\text{in air}} + n_{\text{in depot}}$$

held with zero violations across all 40 replications.

2. **Little’s Law Validation:** On the depot subsystem, the ratio of predicted to observed average number in system was 0.9999, confirming internal model integrity.

Limitations and Sensitivity

- Results assume exponential service times at swap stations, validated against historical data in a separate input analysis.
- Performance is sensitive to drone arrival rate and test-rig processing time. A sensitivity sweep, not shown here, confirmed that the 10.4 s advantage of Policy B remains robust within $\pm 15\%$ of current parameters.
- Adding a second test rig (Policy C) was evaluated but yields diminishing returns compared with the sixth swap station.

Recommendation

Purchase one additional swap station (Policy B).

The improvement is statistically significant at the 5% level, the model has been rigorously verified, and the CRN variance reduction of 1.75 demonstrates that the comparison is reliable. The capital cost of one additional rack is modest relative to the continuous operational benefit of faster drone turnaround during every rehearsal and performance block.

We recommend proceeding with procurement and scheduling installation before the next major rehearsal cycle.

Prepared by Team Mehmet — IE 306 Systems Simulation, May 2026